

AI in the Real World

Data, Power, & Society

CMSE 492 (aka CMSE 101) • Fall 2026

Prof. Danny Caballero

Welcome!



This is not a coding class.

You will not learn to build AI systems.

Instead, you will learn to evaluate, question, and push back on them.

This class aims to help you become a **Critical Pragmatist** when it comes to AI.

Why This CMSE 492 (aka 101)?

By the time you graduate:

- **You will work with AI systems.** Probably with many of them.
- **You will make decisions about AI.** How to use it, when to trust it, whether it should be used at all.
- **Forms of AI will be a part of your everyday life** From websites to software to vehicles to appliances.
- **Others will ask your opinion.** "Does this AI system seem fair to you?" "Should we deploy this?" "Do you trust it?"

This course prepares you to answer those questions with evidence, not intuition.

Who are your instructors?

Prof. Danny Caballero (he/they) - Physics & Astronomy; CMSE; CREATE for STEM

- Lappan-Phillips Professor of Math and Science Education
- Co-direct two research labs (in Physics & Computational Science Education)
- BS Physics, Texas; MS & PhD Physics, Georgia Tech; Postdoc Physics Education, CU-Boulder
- Former custodian, copy shop manager, high school physics teacher in Atlanta Public Schools
- Labor organizer for the Union of Tenure System Faculty (UTSF-MSU)
- Starting my 14th year at MSU (**Go Green!**)

If you are comfortable, please call me "Danny", but Prof. Caballero is fine, too.

Who are your instructors?

Jorge Martinez-Ortiz - CMSE & Political Science

- 4th year PhD Student in CMSE and Political Science
- BS Physics, Baylor University
- International student from Zacatecas, Mexico
- Treasurer for the Comunidad Latinoamericana (CLA) graduate student org.

You can call me “Joy” or “Jorge”.

A Note on the Material

This course covers topics related to AI that may be:

- **Sensitive** (algorithmic discrimination, bias, surveillance)
- **Controversial** (regulation, job displacement, autonomy)
- **Uncertain** (AI's future role, long-term impacts)

It's also going to touch on issues that affect people differently whether that be **gender, race, ethnicity, class, disability, or immigration status**.

- We are all affected by AI and more connected to each other than we realize.
- Each of you has **different biases, experiences, and perspectives**. That includes your instructors.

We will not all agree. That's completely okay. But we will listen and hear with humanity.

Our Shared Framework: DTPA

We use **one vocabulary** to analyze everything.

The Data-Tools-Practices-Actions Framework

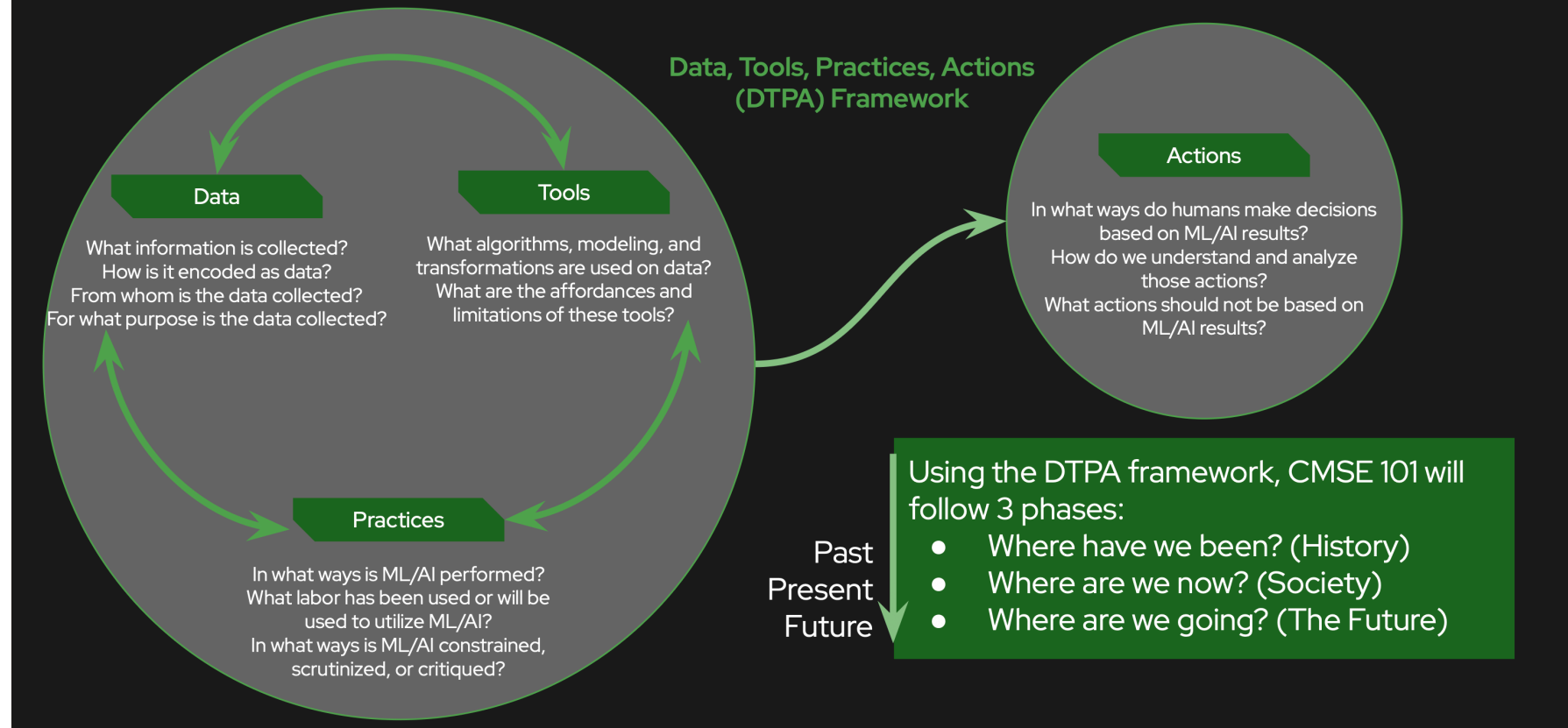
Instead of each person bringing their own lens, we will strive to speak the same language.

This lets us **disagree on conclusions** while **agreeing on facts**.

By the end of the course, you'll be fluent in DTPA analysis. It can be your starting toolkit for evaluating any AI system.

The DTPA Framework

We can understand the major concepts of the CMSE 101 in relation to each other.



The Four Components of DTPA

DATA

What information is collected? Who is included/excluded? What gets erased?

TOOLS

How does the AI actually work? What assumptions are built in? What can it do? What can't it do?

PRACTICES

Who uses this system? How? What labor built it? How is it constrained or enabled?

ACTIONS

What happens as a result? Who benefits? Who is harmed? Does it solve the stated problem?

Course Structure

Three main components grade you:

- 1. **Class Engagement (40%)**
- 2. **Case Studies (30%)**
- 3. **Case Design (30%)**

Your letter grade will be determined by the following cutoffs:

Earned Grade	4.0	3.5	3.0	2.5	2.0	1.5	1.0	0.0
Score Range	> 3.75	3.74... - 3.25	3.24... - 2.75	2.74... - 2.25	2.24... - 1.75	1.74... - 1.25	1.24 - 0.75	< 0.75

1. Class Engagement (40%)

Each component counts equally towards your grade.

- **Daily Class Attendance & Participation**
 - Being attendant is important; we will use lots of group work and being present for your group matters; we will check for your participation daily
- **Forum posts (weekly) + replies (~250 words each)**
 - Posts are expected in response to prompts every week; APA formatting required; Forum replies earn "Deeper Engagement" credit
- **Exit tickets (after every class, ~100 words each)**
 - Each class you will debrief before 5pm; this can also be a place to tell me how you are doing with the material or how things went today

2. Case Studies (30%)

Each component counts equally towards your grade.

- **Case Studies (Google Doc Template)**
 - Analyze AI systems using DTPA each week as a group; turn in your weekly report by Sunday 11:59pm
- **Individual & Metacognitive Report (~250 words)**
 - Reflect on your own learning and effort; bring your values into what we just completed; turn in your report by Sunday 11:59pm

3. Case Design (30%)

Each component counts equally towards your grade.

- **Semester-long group project**
 - Research & design an AI policy or application
- **Two scaffolded assignments (Oct 2nd & Nov 6th)**
 - Support your group's development
 - Receive and respond to feedback from instructors
- **Final showcase (draft due Dec 4th, present last week of class)**
 - Showcase is ungraded but will have class feedback to improve final turn-in
 - Presenting at showcase unlocks final turn-in
- **Final project turn in (Day of our final exam)**

How We Grade: Labor & Standards-Based

This is **not** a traditional grading system.

What it means:

- Your grade reflects your effort and growth, not just "right answers"
 - *There are no right answers to what we are working on*
- Most assignments get a second chance if you don't meet the standard initially
- There's no curve, no "grading on a bell curve"; everyone can earn a 4.0

How We Grade: Labor & Standards-Based

This is **not** a traditional grading system.

What it doesn't mean:

- "Do whatever you want and pass"
- Standards are real, and we hold to them
- You need to revise and engage with feedback

The goal: Learn the material deeply, not just perform well on a test.

We have no tests anyway!

What Does "Meeting the Standard" Look Like?

"In general" - specifics are available in assignments.

Meets Standard (3.0)

- Evidence from reliable sources; documenting sources well
- Thoughtful analysis using the DTPA framework
- Acknowledgment of complexity and multiple perspectives
- Clear and respectful communication

Exceeds Standard (3.5–4.0)

- Deep research beyond assigned readings; new resources and perspectives
- Metacognitive reflection on your own learning; deep engagement
- Leadership, Exceeding Efforts, or Novel Contributions in group work

Class Culture & Expectations

Here's what we need from you:

- **Show up.** Physically and mentally. Exit tickets require attendance.
- **Engage thoughtfully and respectfully.** Disagreements are ok, arguments are not.
- **Think about your position.** No one has your experience and background.
- **Disclose AI use.** If you used it tell us. We trust you.
- **Meet deadlines.** Forum posts due before class each week. Most of other is due Sundays 11:59pm.
- **Ask questions.** If something doesn't make sense, say so.

Class Culture & Expectations

Here's what you should expect from us:

- Clear expectations and timely feedback
- A second chance on most assignments
- Support if you're struggling; space if you need it
- Respect for your time and perspectives; valuing you as a person
- Stepping in to manage issues, concerns, and disagreements

This Week's Agenda

1. **Course intro** (you're here ✓)
2. **Group the class!** - Get up and move around!
3. **Constructing Categories** - How we perceive people matters.
4. **Wednesday: Who do we make into groups?** – We'll work through the Census timeline together
 - *In your assigned groups*
5. **Wednesday & Friday: First Case - Census Data** – Focusing on the Data in DTPA

Exit ticket – Your first one is due by 5pm today on D2L.

Making Groups - Part 1

- Break into six random groups of roughly equal size
- You have five minutes to come up with as many things you all have in common
 - Nothing pejorative (e.g., *we all hate Drake*) or physics (e.g., *we all have red hair*)
- You will have three more minutes to identify the top 2-3 things you all think the rest of the room **won't** share
 - Each of you write on those 2-3 things on index cards (1 idea per card)

Making Groups - Part 2

- Shuffle into six *new* random groups of roughly equal size
 - No more than one other person from your first group
- Share your 2-3 things; is there anything that hangs you all together?
 - Again, nothing pejorative (e.g., *we all hate Drake*) or physics (e.g., *we all have red hair*)
 - Decide on top 2-3 (discard unused cards; write new cards)
If you can't find 2-3 things, shuffle again!
- **Once you 2-3 things with a new group, try to name your group!**

Making Groups - Wrap Up

Share out the commonalities!

Discussion:

- Are these categories the same "kind of thing"?
- Are they truly unique?
- What does it mean to categorize people?

Our Kind of People - Bayeté Ross Smith



Source: <https://www.bayeterossmith.com/our-kind-of-people>

Our Kind of People - Bayeté Ross Smith

- Break into six groups of roughly equal size; we will count off group numbers
- On D2L, open the link to the [Gallery: Our Kind of People](#) Activity under In Class Materials -> Day 01.
 - Write your names at the top of your Group's Tab in the Google Doc

Read the instructions and start working (7 mins.)

Write down a list of labels or assumptions someone might make about the person in the photo, answering the question: What assumptions might someone make (regardless of what this person intended to convey) about this person's identity?

Reflection Questions (10 min)

- Why did each group have different labels for photos of the same person?
- How do we use labels to understand each other? When might those labels be incorrect or incomplete?
- Where do the labels and stereotypes we apply to others come from?
- How does clothing allow people to emphasize certain parts of their identities? In what ways does it allow people to hide other aspects of who they are?
- These images are part of a larger work by an artist named Bayeté Ross Smith, titled “Our Kind of People.” What message do you think the artist was trying to convey by creating this project?

Reminders for this week

- Don't forget to complete today's Exit Ticket by 5pm (on D2L)
- Review your group assignment on D2L and connect with your group before Wednesday's class!
- Spend time before Friday (~90 minutes) going over this week's preparatory materials

Questions?

Email Danny (caball14@msu.edu) and/or Jorge (martjor@msu.edu)

Have a great first week!